

Spectroscopic Survey of Small Solar System Bodies with Ariel

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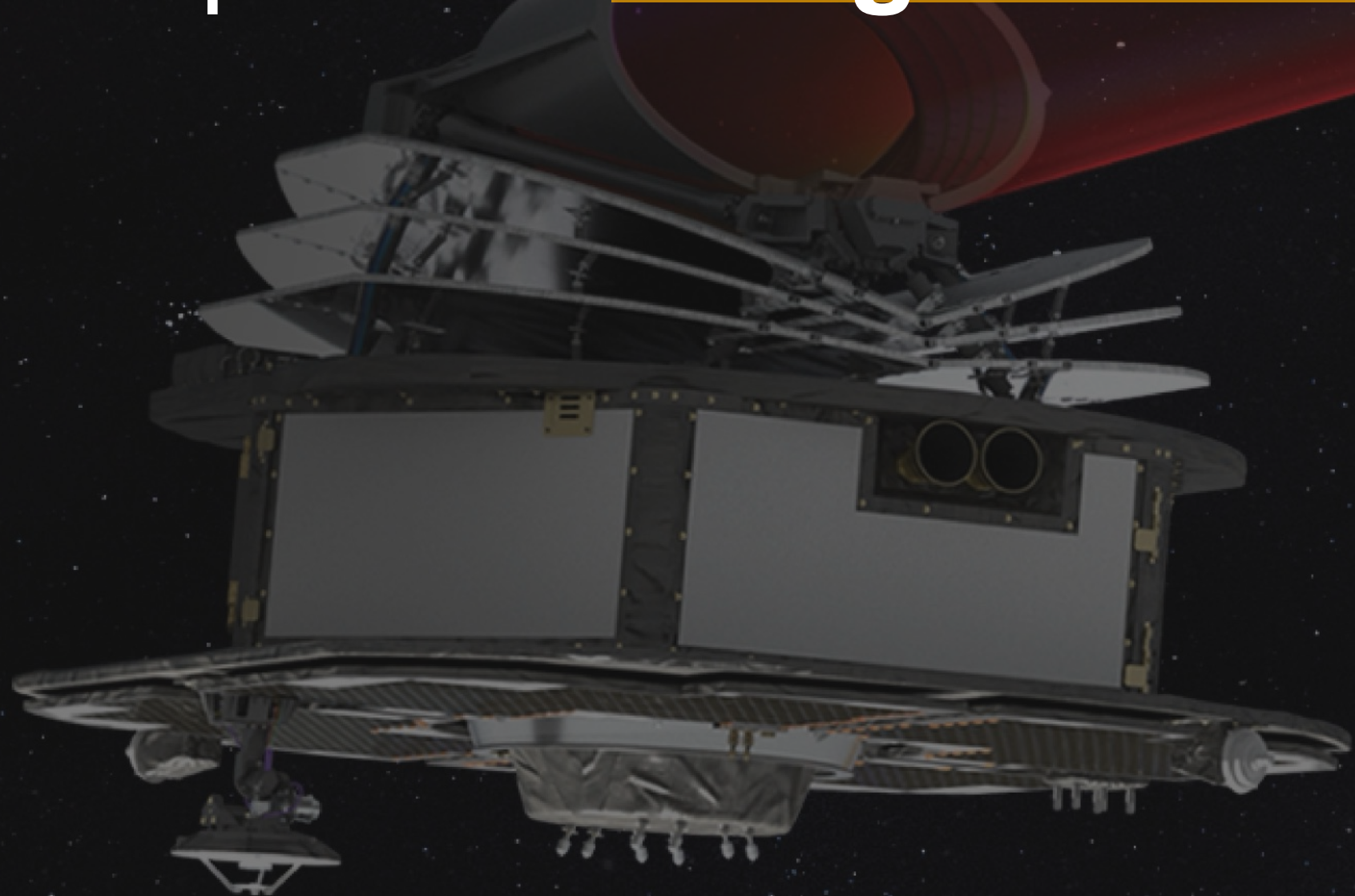


September 16, 2026 13:55–14:10 — Ariel Consortium meeting 2026

Credit: ESA

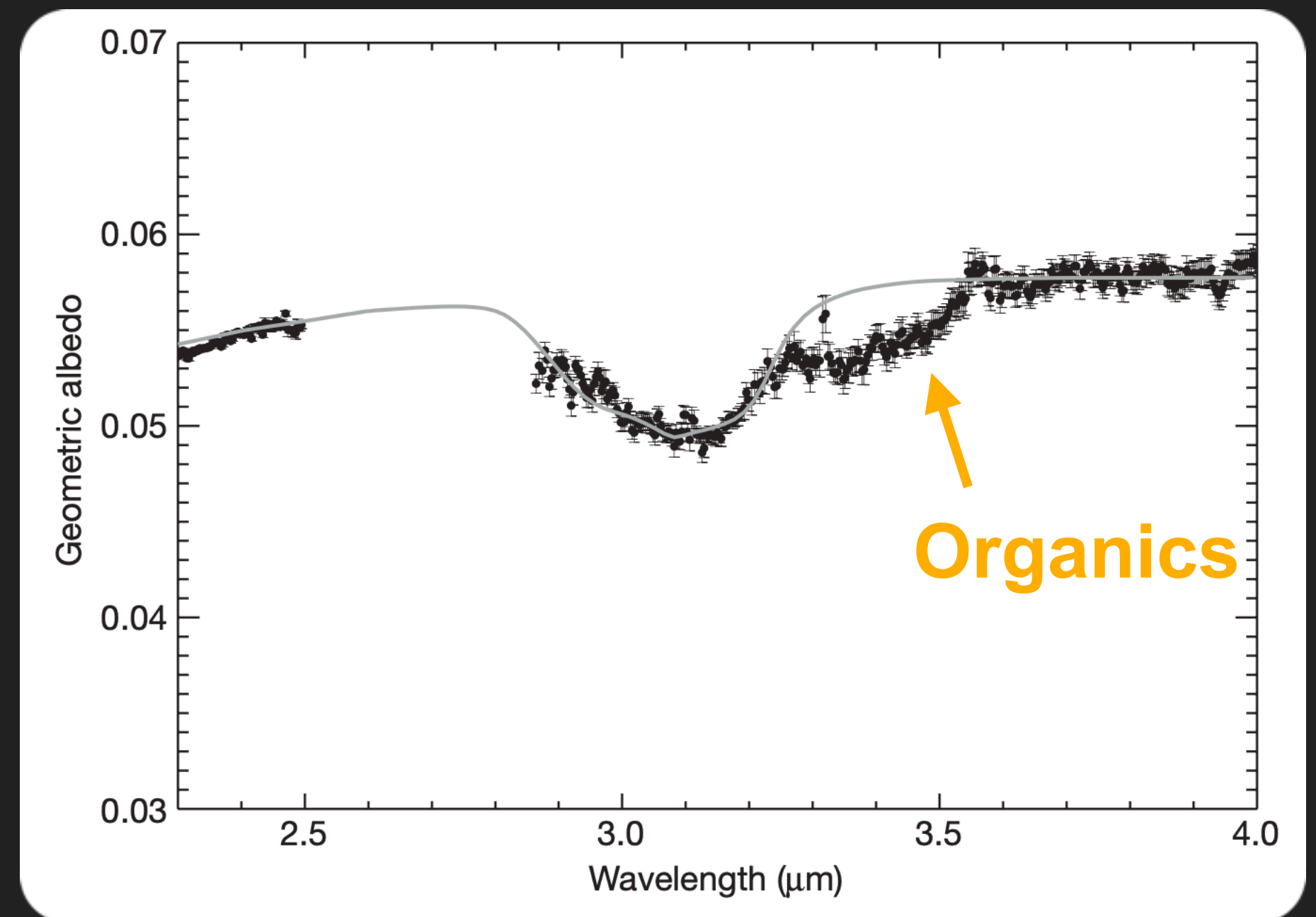
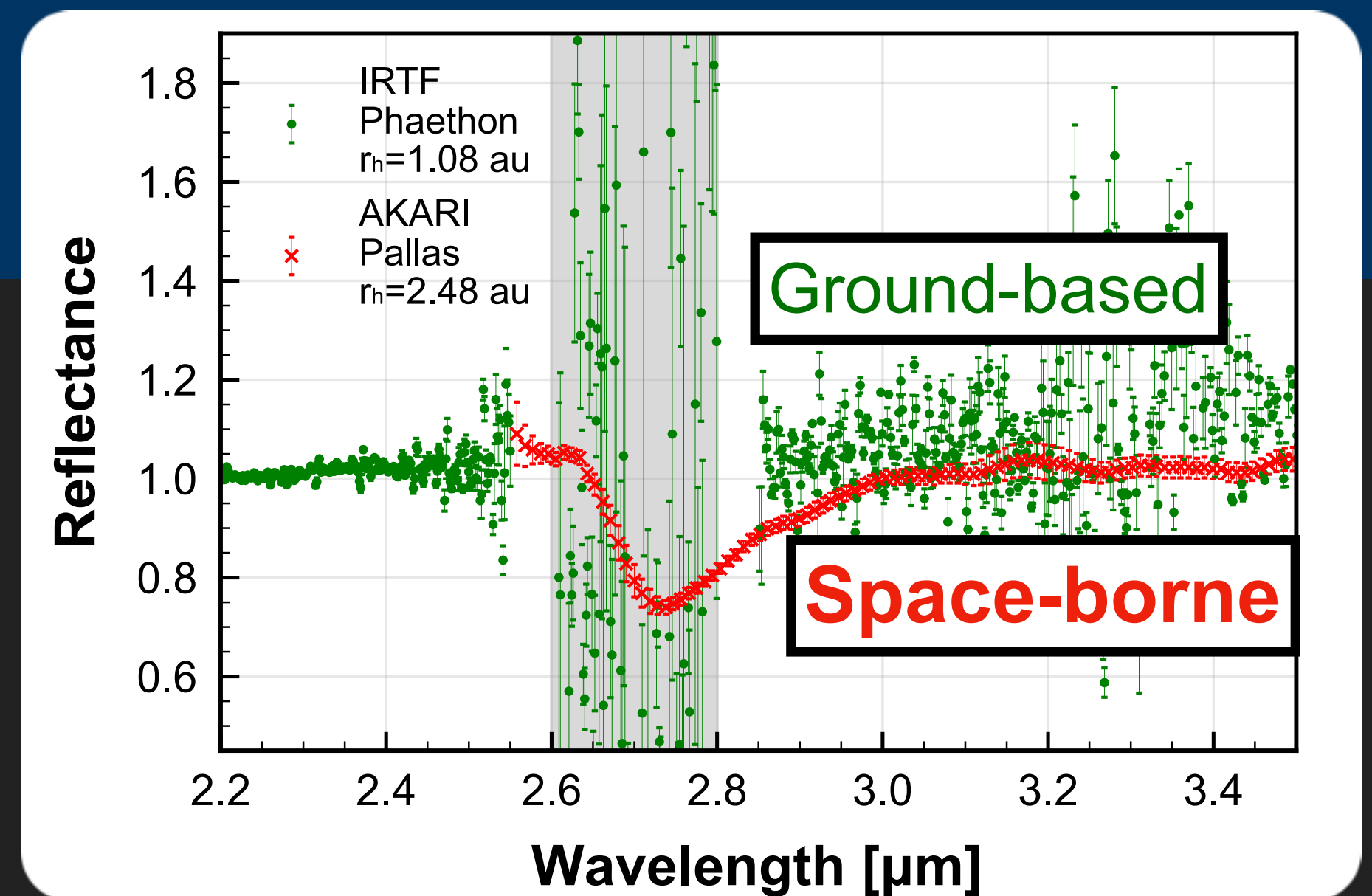
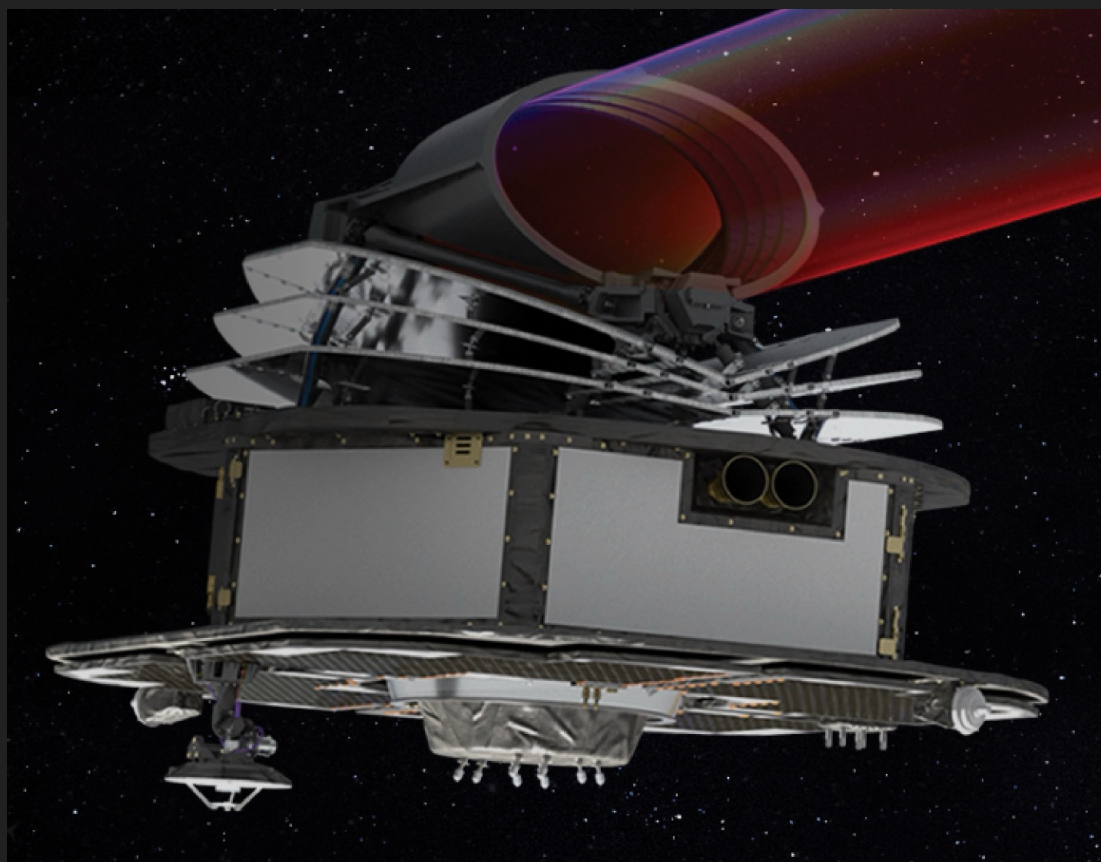
Takeaways

- Ariel offers a unique opportunity to explore hydrated materials and organics across the Solar System.
- Hundreds of accessible asteroids provide a large and flexible target pool for complementary science.



Scientific motivation

- Small Solar System bodies (SSSBs) preserve the history of our Solar System
- The 3 μm region reveals
 - ▶ Hydrated materials
 - ▶ Organics
- Ariel (0.5–7.8 μm):
unique window into this spectral region



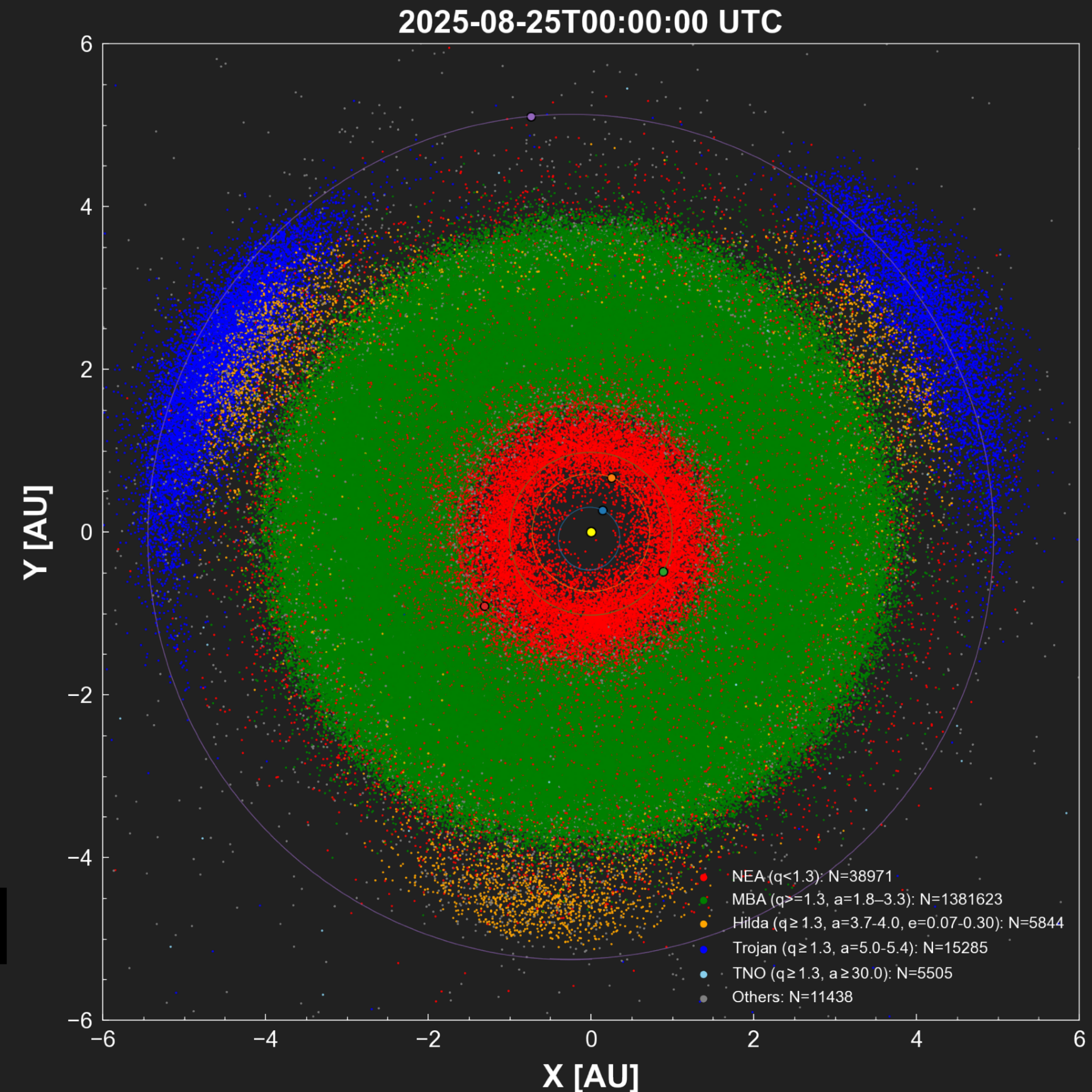
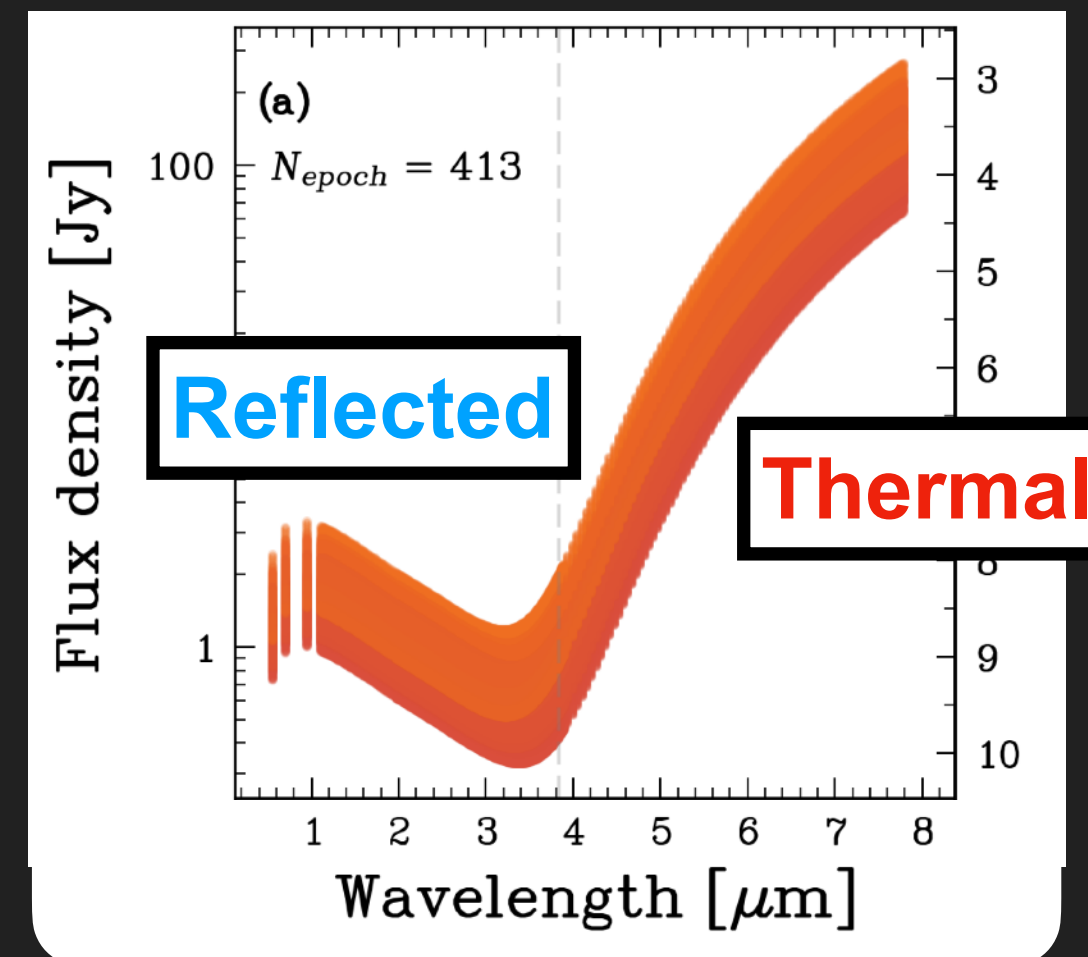
Method

1. Visibility

- ▶ NASA JPL Horizons
 - 2032-01-01–2035-12-31 (timestep: 1 day, from *Gaia*)
 - 1,458,430 SSSBs
- ▶ Requirements
 - $V < 13.6$ (nominal) and $V < 16.0$ (extended)
 - $65^\circ < \text{Solar elongation} < 115^\circ$

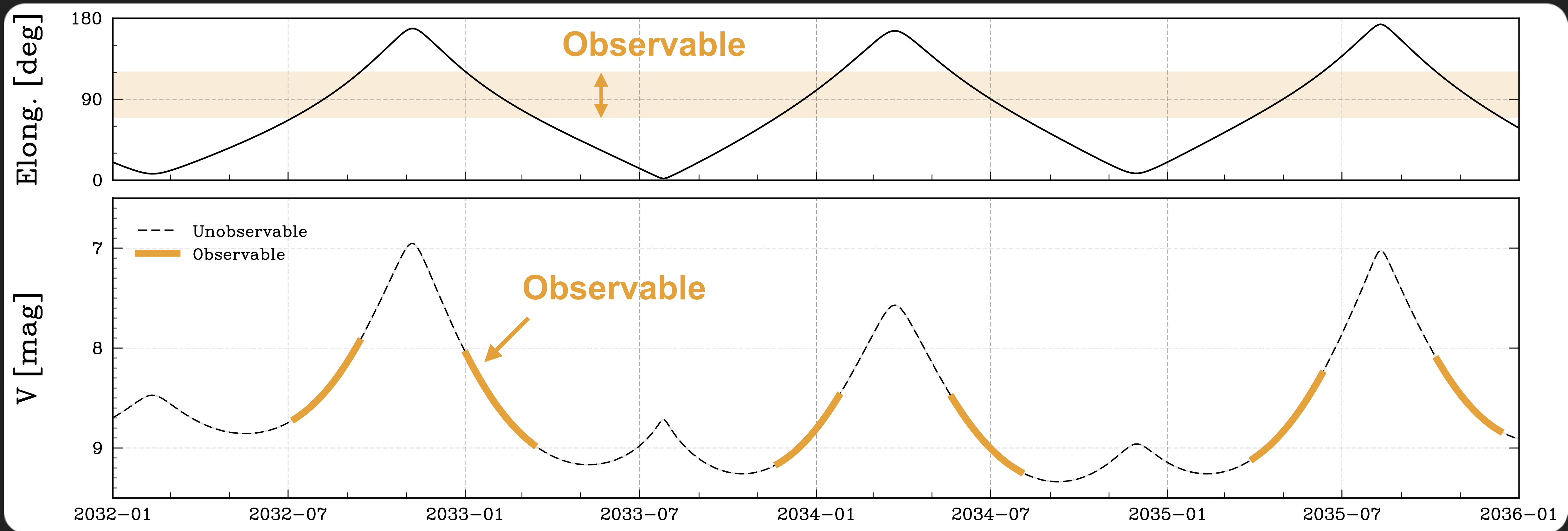
2. SED

- ▶ Reflected light + Thermal emission
 - Predicted V band mag + spectral type & albedo
 - Standard Thermal Model (STM)



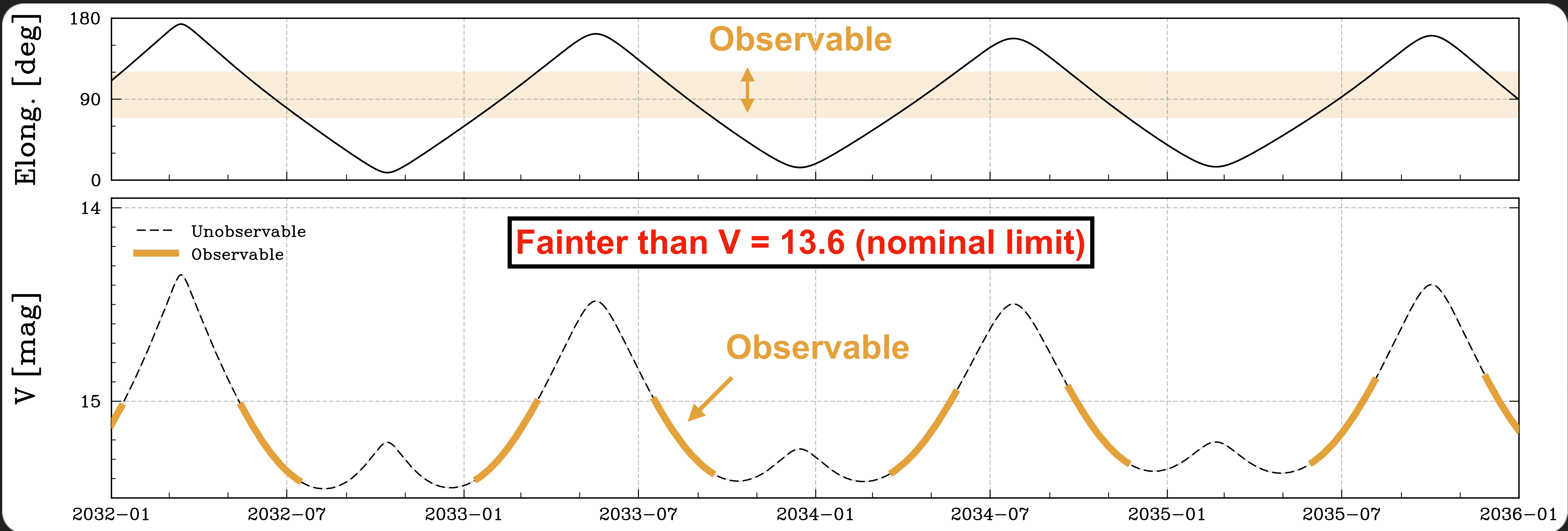
Result: Visibility

- Bright case: (1) Ceres in Main Belt



Result: Visibility

- Faint case: (624) Hektor

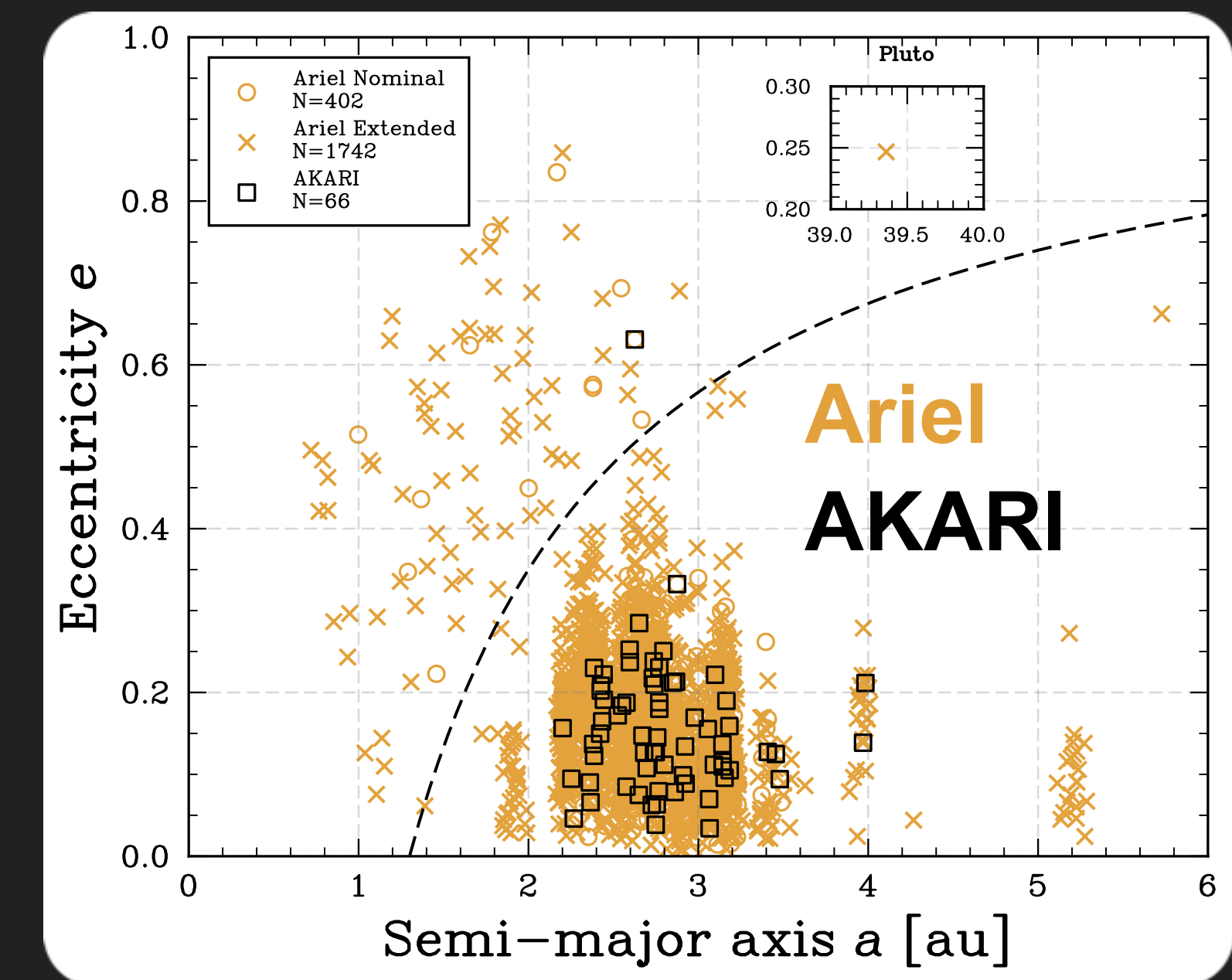
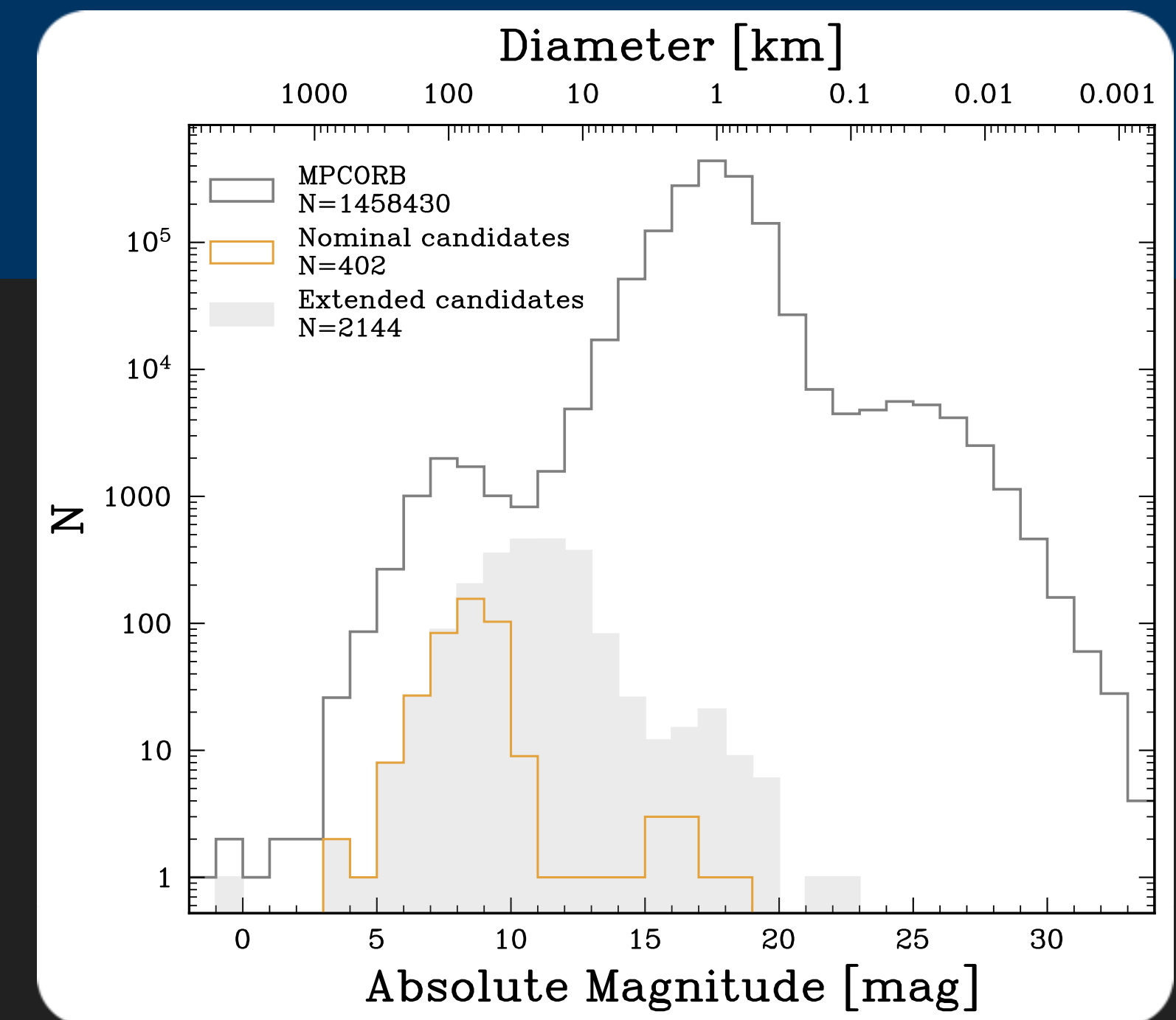
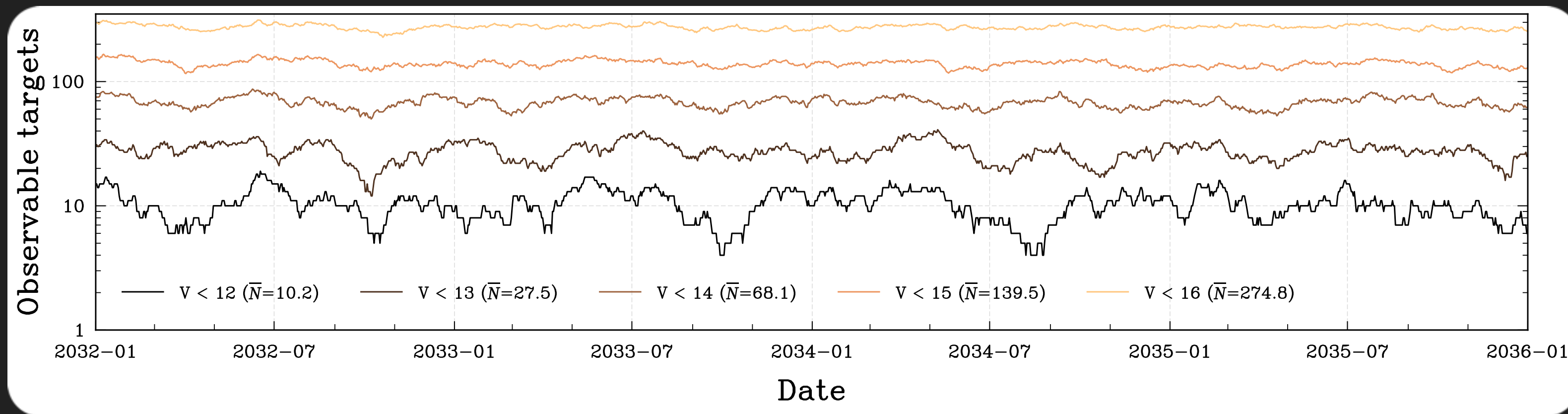


Result: Visibility

- **Candidates**

- ▶ $N=402$ ($V < 13.6$)
- ▶ $N=2144$ ($V < 16.0$)
- ▶ Diverse population (orbits, size, composition)

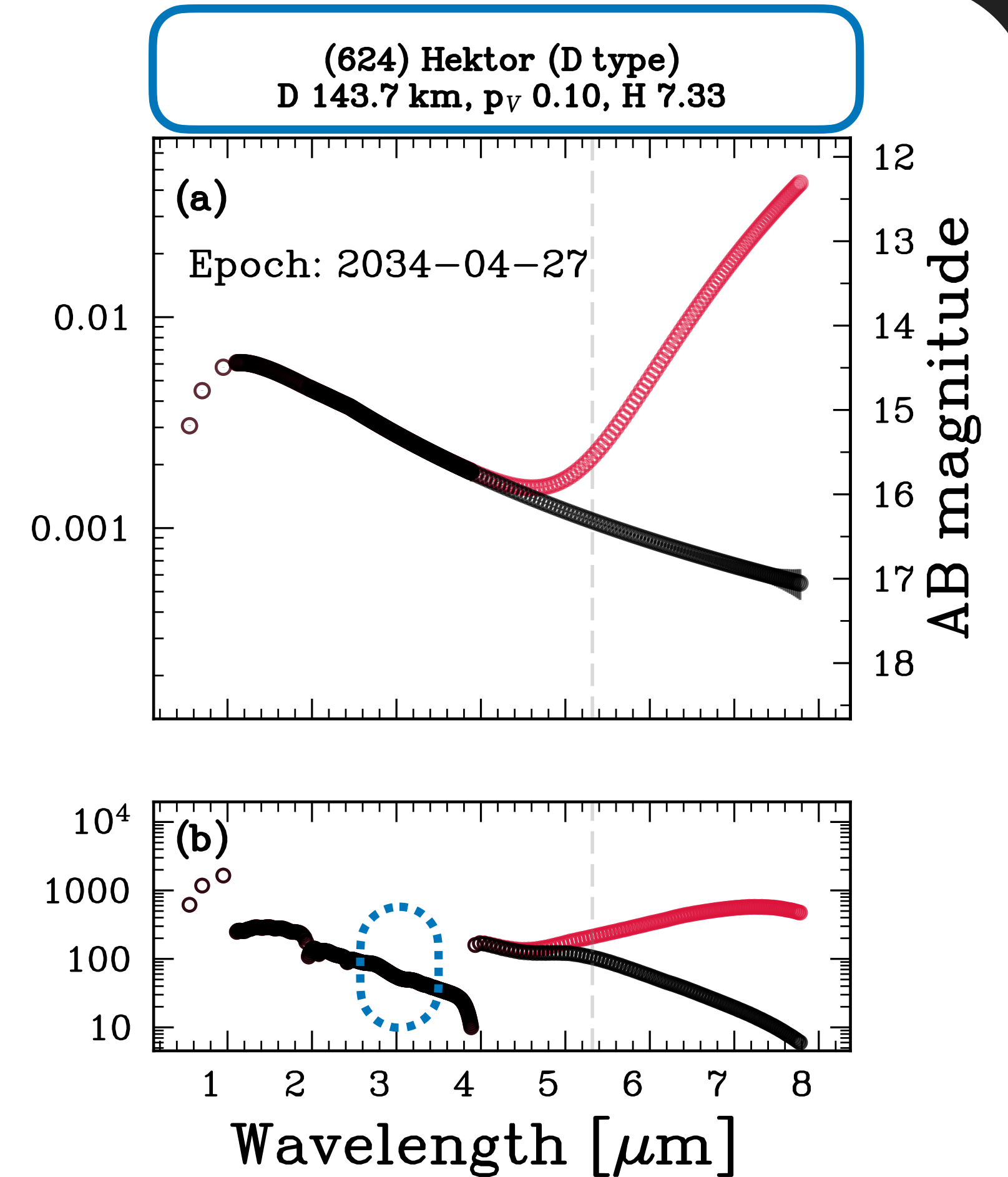
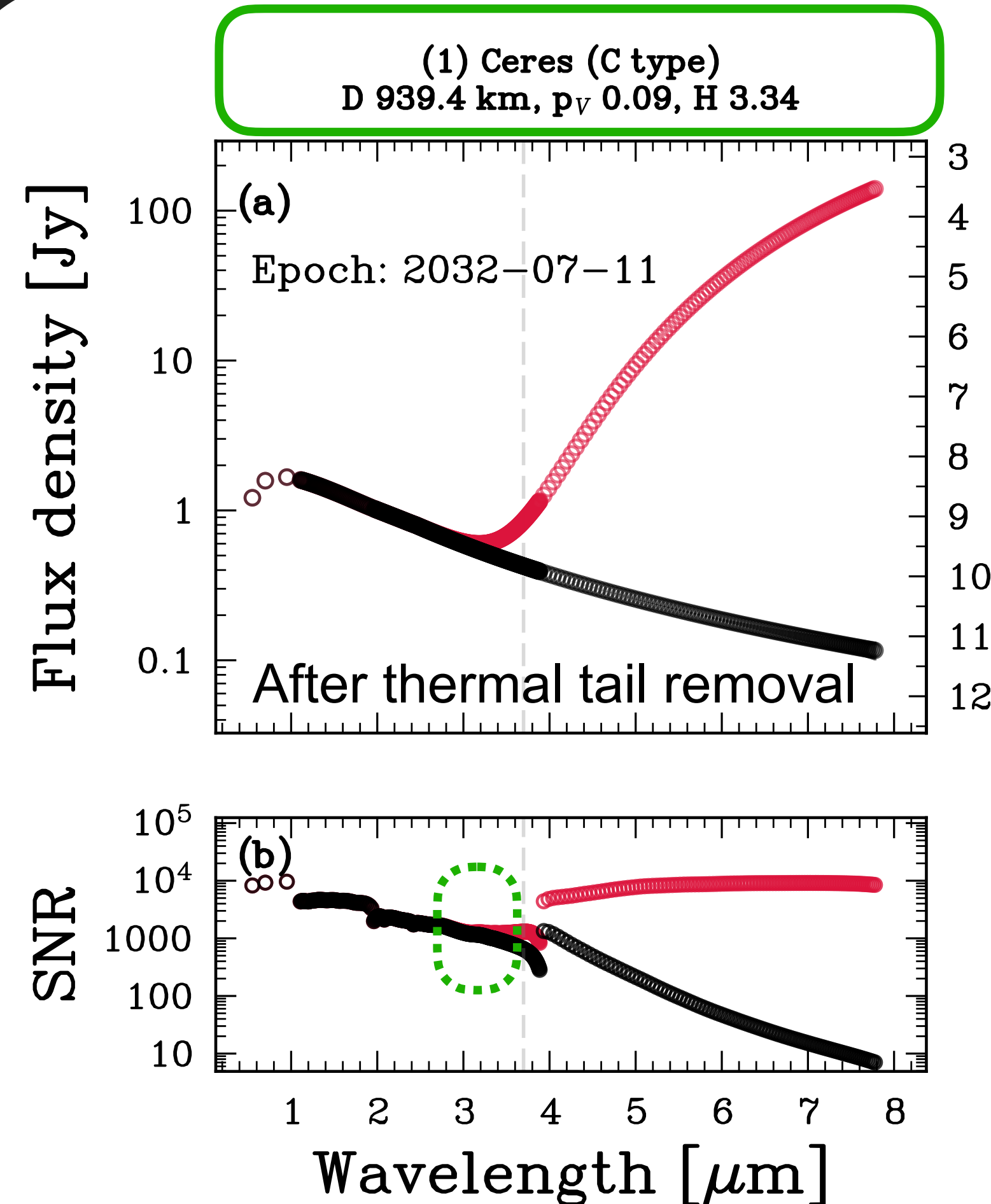
- $N \sim 10$ ($V < 12$), $N \sim 30$ ($V < 13$), $N \sim 70$ ($V < 14$)
at any time, across the sky



Result: SED

- Nominal targets:
 - $S/N > 100$ at $\sim 3 \mu\text{m}$

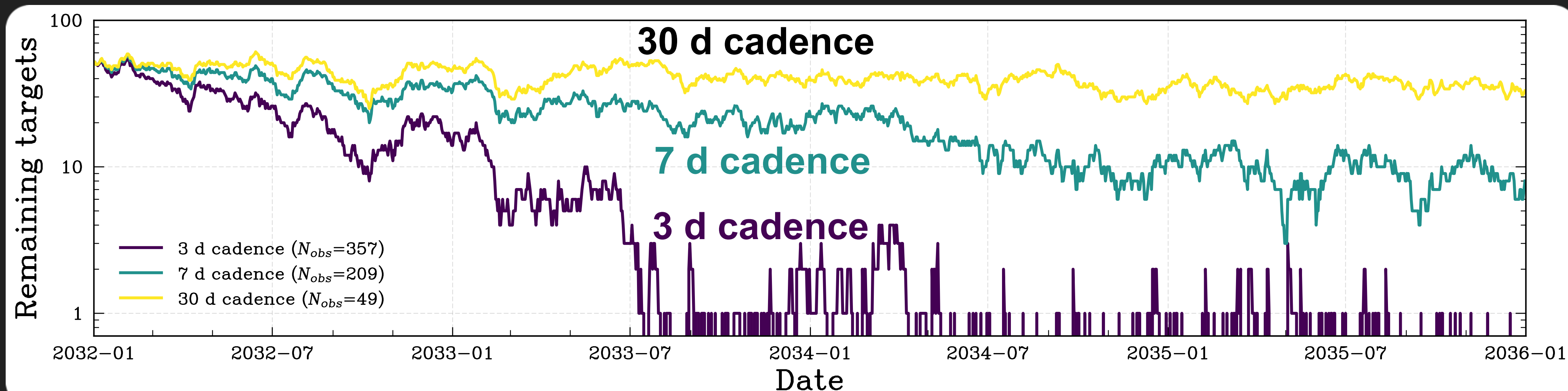
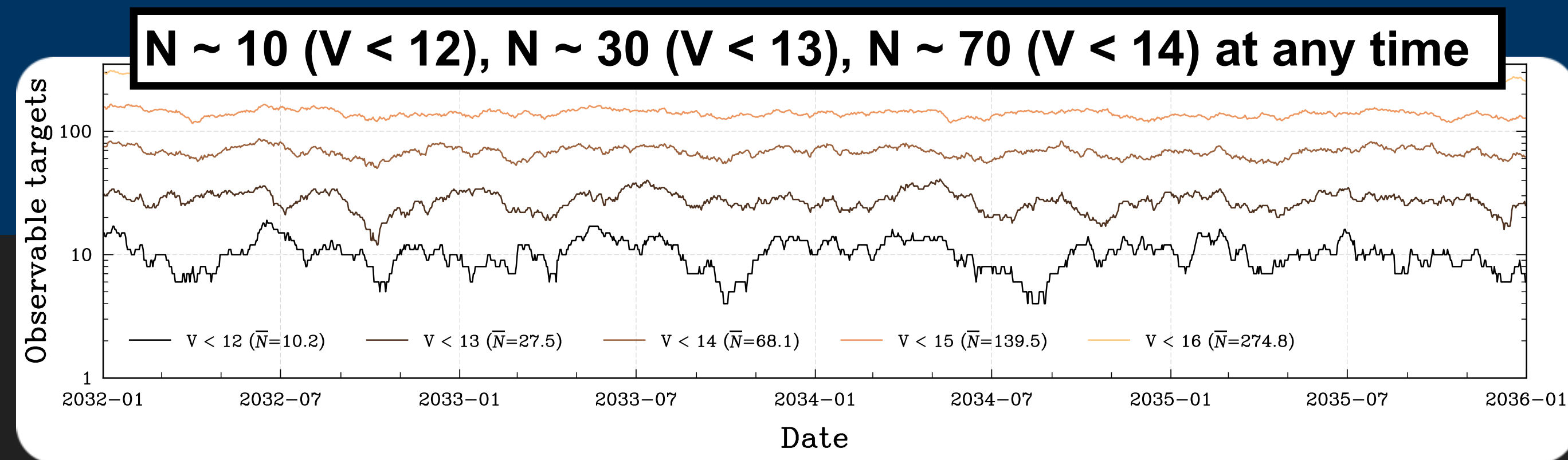
- Extended targets:
 - $S/N < 100$ at $\sim 3 \mu\text{m}$ but still useful



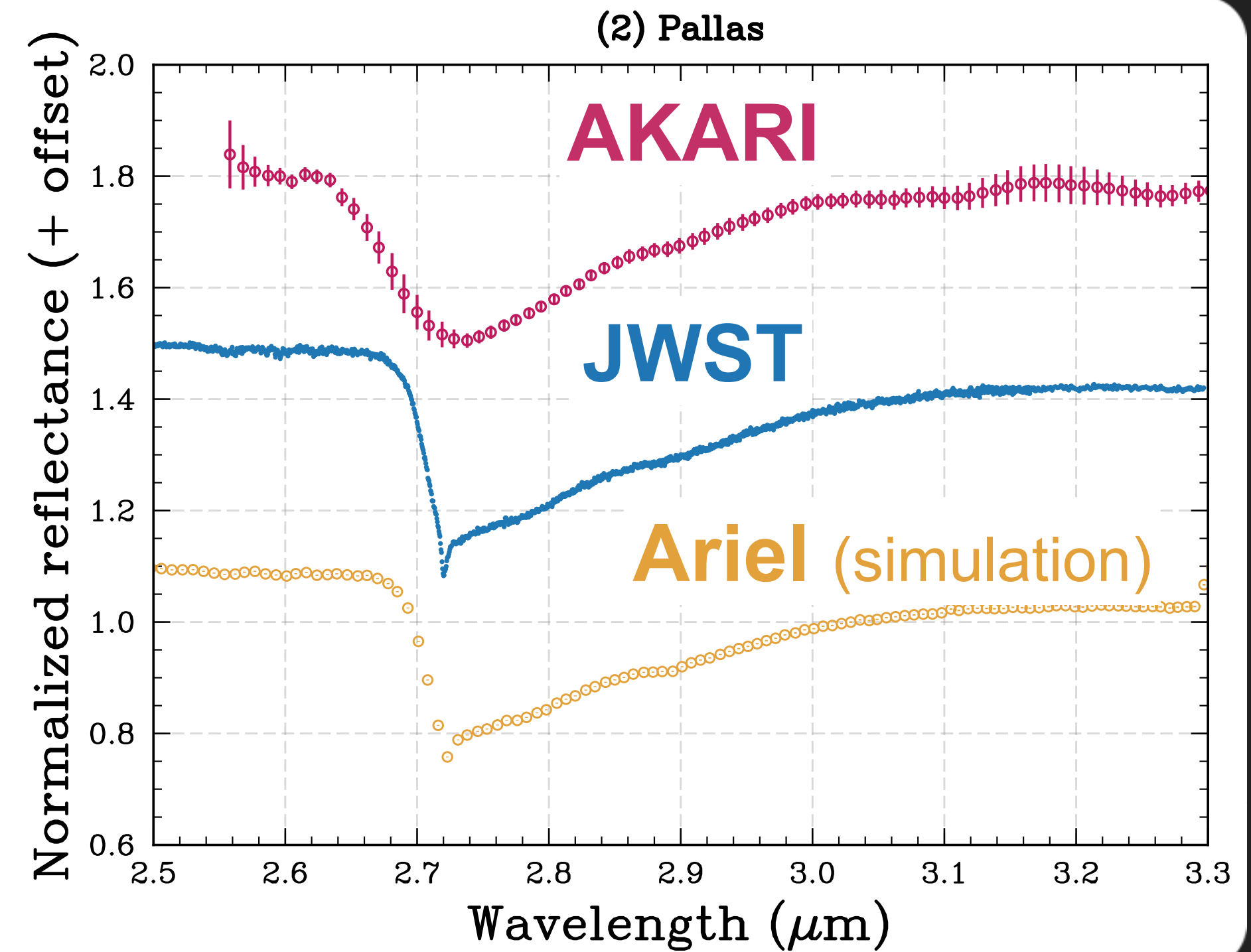
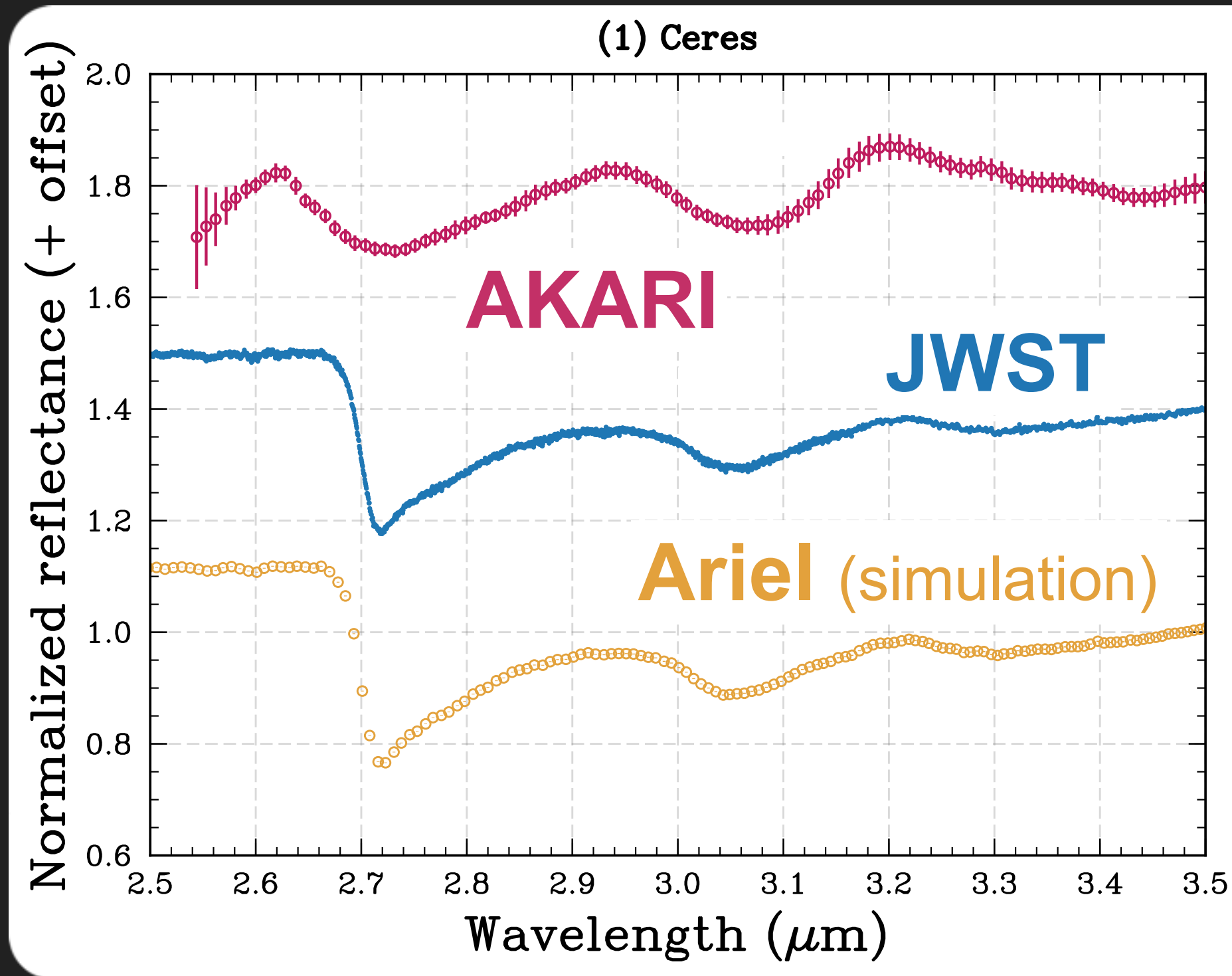
Flexible Target Pool

- **Mock observations**

- ▶ 3, 7, 30 days cadance
- ▶ Nominal case ($V < 13.6$):
N=357, 209, 49 unique samples
- ▶ Extended case ($V < 16.0$):
Pool is never empty



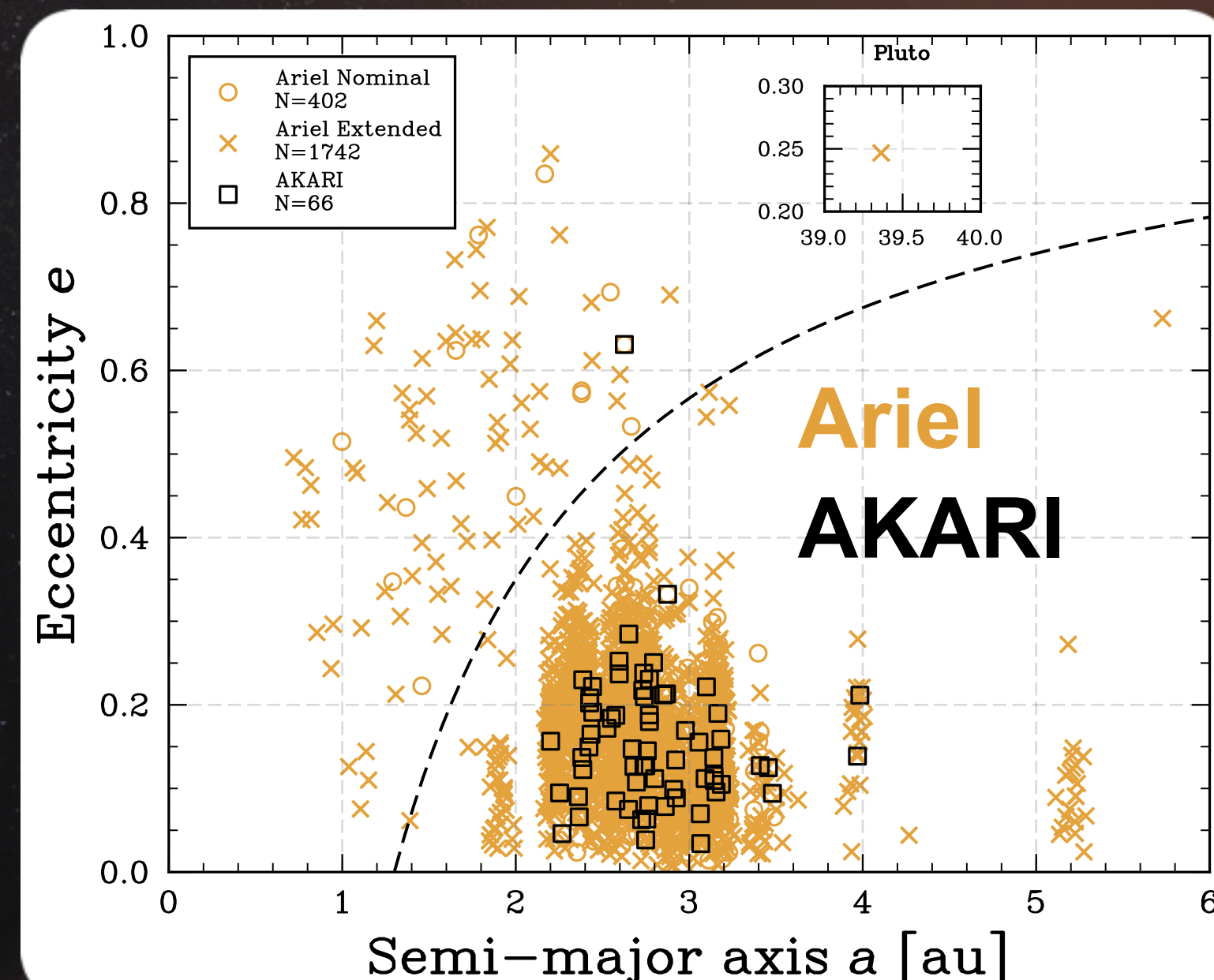
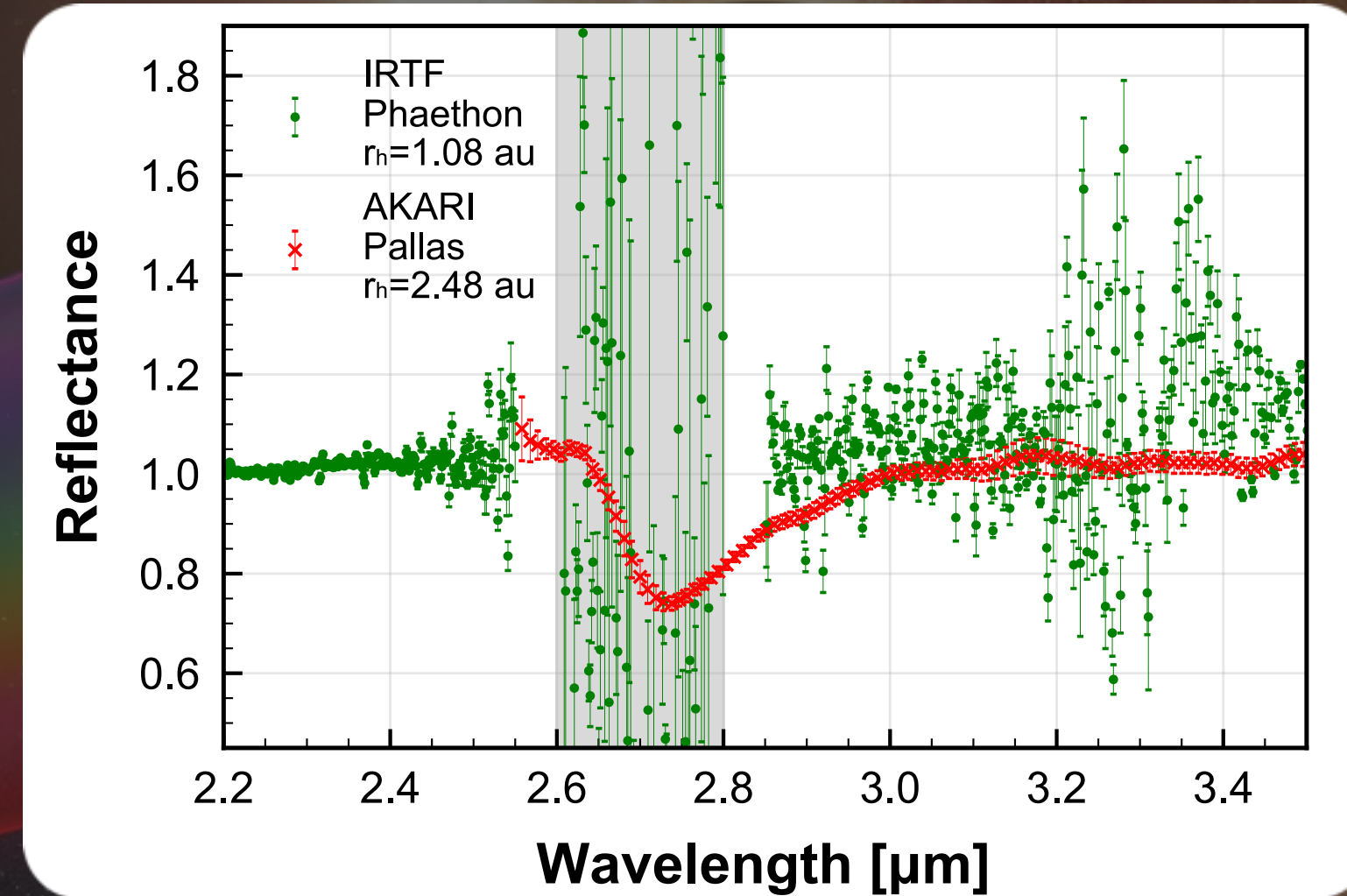
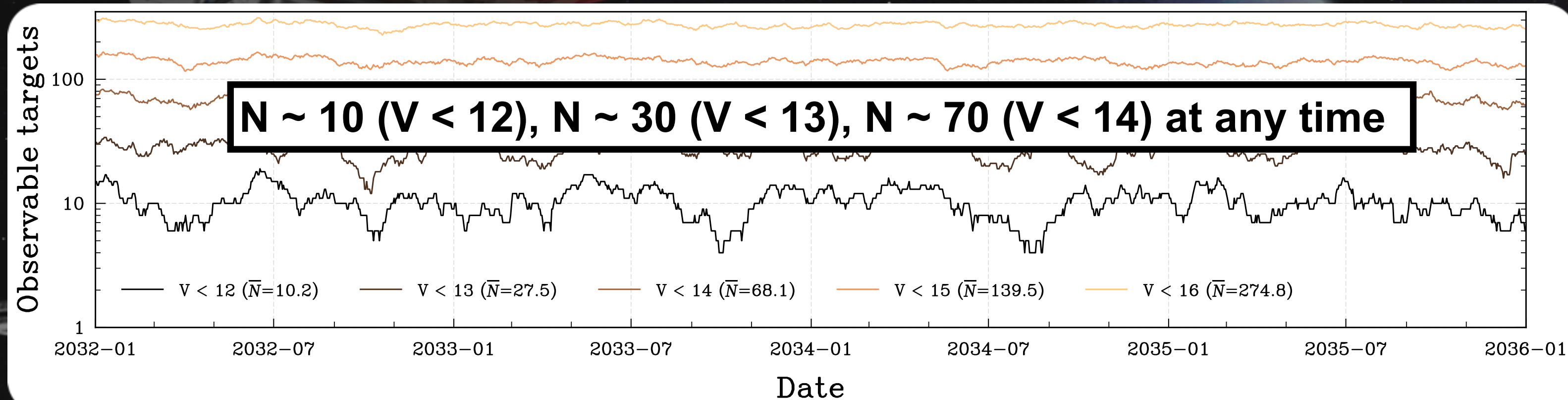
Unique Survey of Small Bodies



	AKARI/IRC	JWST/NIRSpec	Ariel
Wavelength [μm]	2.5–5.0	0.6–5.3	0.50–7.8
Resolving power at 3 μm	100	2700	140
Number	66	O (10)	~ 50–350

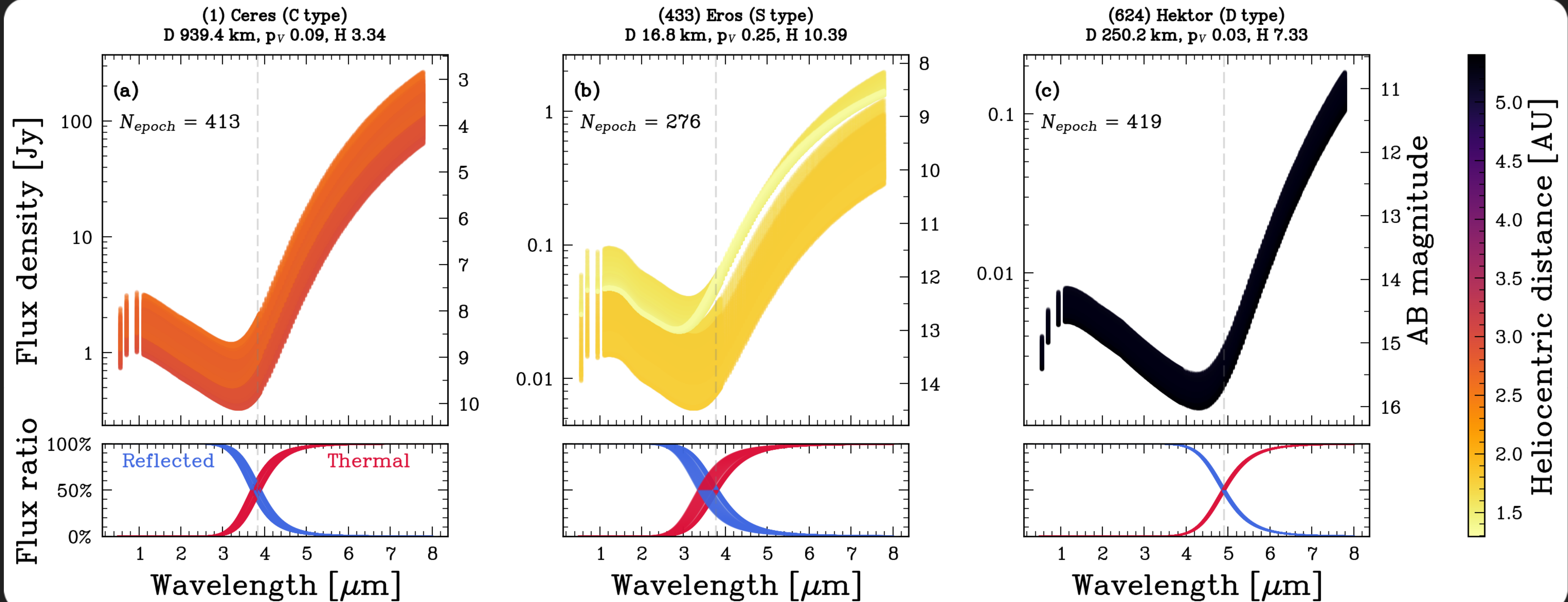
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Back up

SED



Candidates

